

GYM LEADER LAB | AI PROGRESS UPDATE

Smarter bosses, still fair.

The next AI plan turns existing battle infrastructure into more human-feeling decisions without broad search or hidden player information.

Buildable

Public-info AI

Plan only

Fair pressure, not clairvoyance



The project is in a credible AI iteration phase.

The current ROM already builds and already has a split boss AI overlay. The plan is not a rewrite. It is a sequence of small, audited heuristics that improve move quality while preserving fairness.

Verified now

Gold/Silver build passed

Normal build produced pokegold.gbc and pokesilver.gbc during the planning pass.

Implemented foundation

Boss AI overlay is live

Move scoring, switching, public threat masks, lookahead, and weighted best-vs-second selection already exist.

Next milestone

One narrow patch first

Public revealed-spinner hazard retention is the first planned implementation; it is not implemented yet.

The AI stack is already more than vanilla trainer behavior.

Public state

Visible active Pokemon, revealed moves, seen species, HP/status/boosts, hazards, observed switches.



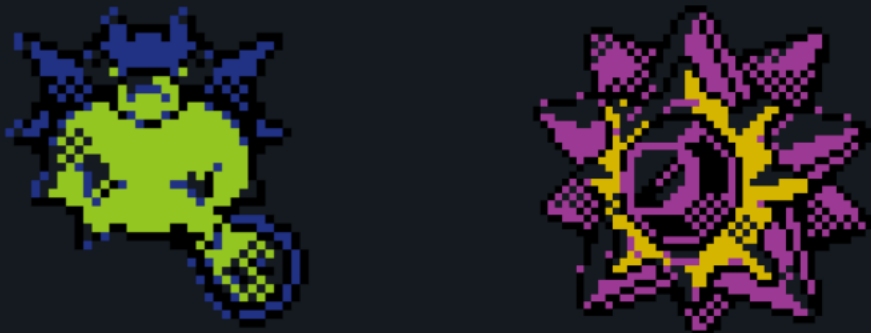
Boss scorer

KO pressure, deny-KO, utility fail gates, plan/role bias, setup pressure, phazing, Spikes, Rapid Spin.



Selection and trace

Lookahead for near-top moves, weighted best-vs-second selection, trace fields under BOSS_AI_TRACE.

Two pixelated images of a boss, likely a Dragon King, are shown side-by-side. The left image is labeled 'Boss move set' and the right image is labeled 'Player public state'. Both images show the boss in a dynamic pose with multiple limbs and a central eye-like feature.

Boss move set

Player public state

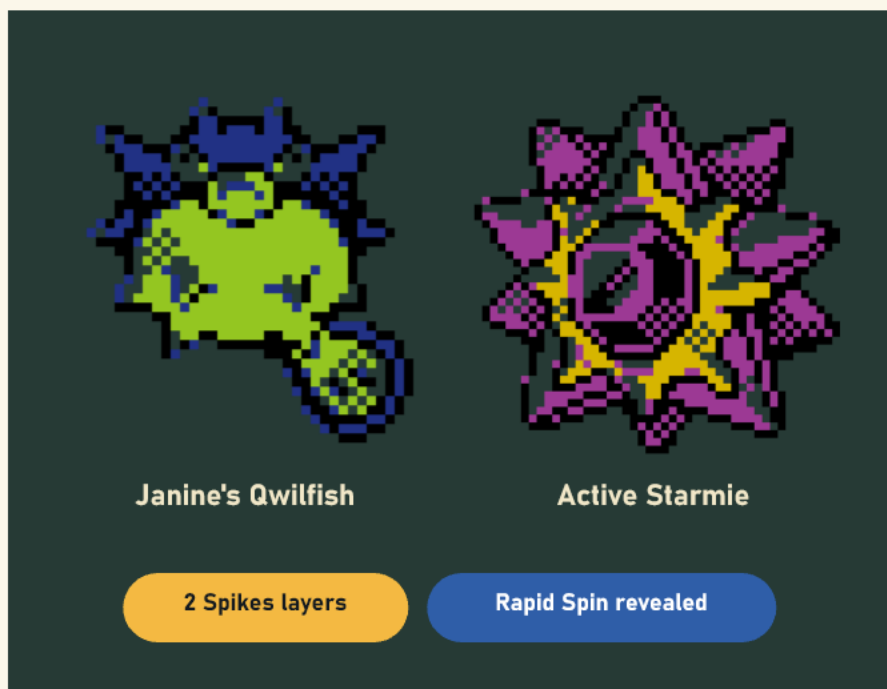
Current selector already mixes close best-vs-second choices; the plan tightens what counts as a defensible choice.

The work shifts difficulty from raw stats to readable decisions.

	Before: current or identified gap	After: planned behavior
Hazards	Layer-aware Spikes exists, but no dedicated active revealed-Rapid-Spin penalty was found in the Spikes branch.	Stack only when it still converts; otherwise pressure or cash out against the revealed spinner.
Setup	Existing pressure checks reduce bad setup, but the plan calls for tighter low-HP public-KO cases.	Setup is suppressed unless it immediately wins, survives, or preserves the route.
Sacrifice	Self-KO moves already have risk handling, but route-value discipline is a later target.	Explosion-style trades fire when they remove a public route piece, not for low-value chip.
Predictability	Weighted best-vs-second already exists; broad new randomness is not the first patch.	More close choices become human-plausible only after hard rejects keep bad moves out.

No row represents shipped behavior unless marked current. Planned rows require implementation and validation.

Hazards become smarter when Rapid Spin has been revealed.



Before

No dedicated check in `.ApplySpikesLayerBias` for active revealed Rapid Spin. The score mainly follows layer count, pressure, and switch chance.

After the first planned patch

If Rapid Spin is in the active player's revealed used-move list, extra Spikes get discouraged. Hidden spinners are not inferred.

Expected player-facing effect: Janine and Koga stop looking scripted around hazard loops and start protecting hazard value fairly.

Setup decisions become less exploitable under public lethal pressure.

Current foundation

The scorer already considers setup under pressure.

The roadmap tightens narrow cases where visible HP bands and public KO pressure say a boost turn is likely dead unless it immediately changes the route.



This is a later roadmap item, not the first implementation patch.

Before	A setup move can still be a tempting scorer result in edge cases if the immediate public danger is underpriced.	Planned gate: low-HP setup loses unless setup wins, survives, or preserves a route.
Player read	The player can sometimes exploit passive boost turns by attacking through them.	The boss uses the public board to choose damage, denial, or preservation sooner.
Fairness	No private damage roll or hidden move slot is needed.	Uses visible HP/status, public threat masks, revealed moves, and boss-owned state only.

Sacrifice moves become route tools instead of spectacle.

The plan treats Explosion and Selfdestruct as human trade decisions.

Current gap	Sacrifice moves can be over- or under-valued without a route-value gate.	Planned scoring: spend them only when public route value is high.
Route value	Meaningful trades remove a public threat, deny setup, or open the boss's route.	Measured through fixtures before source changes.
Guardrail	Do not call hidden party or current-input helpers to decide whether the trade is good.	Use active visible pressure and already seen species only.

Roadmap item after hazard retention, four-move saturation review, and anti-dead-setup tightening.



The deck does not claim Elo. It claims measured gates.

Green in the planning pass

● Gold/Silver normal build	PASS	Fresh build completed.
● No-cheat audit	PASS	Ordinary boss AI boundary currently clean.
● Debugger regression	PASS	43/43 strict pairwise labels agreed.
● Preference validation	PASS	57 fixtures validated.

Must refresh before implementation

● Memory budget audit	BLOCKED	dev_index.md is stale after the build.
● Live trace ledger	BLOCKED	Trace ROM hash mismatch.
● ROM hash compare	NOT A CLEAN GATE	Normal Gold/Silver hashes diverge from baseline.
● Haki/oracle	QUARANTINED	Allowed only under its documented exception contract.

This slide reports actual tool outcomes from the planning pass; no win-rate or 1500 Elo claim is made.

The path is deliberately small, reversible, and evidence-led.



Leader identity improves
through scoring, not bespoke
scripts for every fight.

● 0. Reset the floor

● 1. Revealed-spinner hazards

● 2. Four-move saturation

● 3. Anti-dead setup

● 4. Cash-out discipline

FIRST

FIRST PATCH

NEXT

NEXT

LATER

Regenerate dev_index.md and refresh trace hashes or document deferral.

0 new WRAM; use exact active revealed Rapid Spin only.

Stop respecting speculative coverage after four real moves are public, after edge-case audit.

Suppress low-HP setup under public KO pressure unless it changes the route.

Make Explosion-style trades route-aware and measurable.

The plan explicitly defers broad search, large counters, and personality tables until smaller heuristics are proven.

The improvement is product-visible, not just technical.

After implementation, bosses should feel more prepared because they punish public board mistakes, preserve important pieces, and vary close good choices.

The credibility comes from the constraints: buildable ROM, byte-aware patches, no hidden player information, and scenario gates before claims.

Fairer difficulty

More replayable bosses

Measured rollout



Next concrete milestone: implement Public revealed-spinner hazard retention with three fixtures, normal build, map delta, and audit floor.